

ARTI 404 - Image Processing

Lab 4 Report

Intensity Transformations and Filtering - Spatial Domain

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Objective: implement and analyze thresholding, contrast stretching, histogram equalization, and histogram matching using Python and scikit-image.

Contents

1	Lab Overview	1
2	Guided Tasks	1
3	Assessment Task 1 - Contrast Stretching	2
4	Assessment Task 2 - Histogram Equalization	2
5	Assessment Task 3 - Histogram Matching	3
6	Final Results Summary	4
7	Resources	4
8	Submission Contents	4

1. Lab Overview

This report documents the completed activities for Lab 4. The laboratory manual identifies the topic as Intensity Transformations and Filtering in the Spatial Domain and specifies Python 3 with Jupyter/Spyder and image-processing libraries including NumPy, OpenCV, Pillow, scikit-image, and Matplotlib.

For the assessed work, the notebook uses the built-in scikit-image images required by the manual: **moon** for Tasks 1 and 2, **Chelsea** as the source image for Task 3, and **rocket** as the reference image for Task 3.

Guided thresholding was also implemented as part of the lab. Since the manual example references an external Parrot image that was not included with the provided manual, the notebook uses scikit-image's built-in **camera** image to demonstrate the same fixed-threshold operation.

2. Guided Tasks

2.1 Fixed Thresholding

Fixed thresholding converts a grayscale image into a binary representation. For threshold T , pixels above T are displayed as white and the remaining pixels as black. The notebook evaluates $T = 0, 50, 100, 150,$ and 200 .

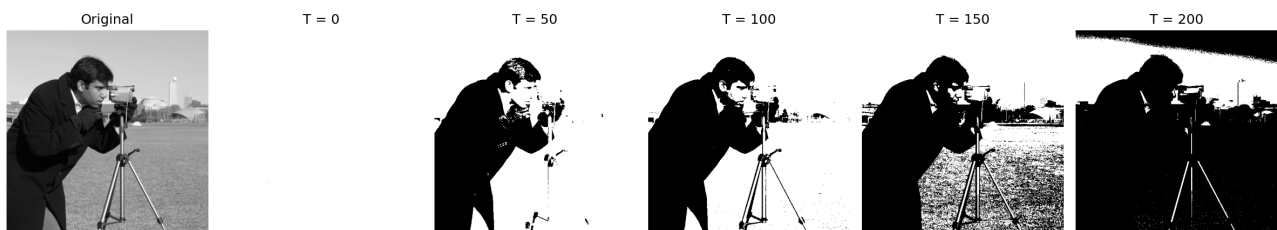


Figure 1. Fixed-threshold results for the guided thresholding task.

2.2 Contrast Stretching - Guided Example

Following the procedure in the manual/resource example, the moon image was contrast-stretched using the 2nd and 98th percentiles. For this image, the percentile values were **78** and **129**.

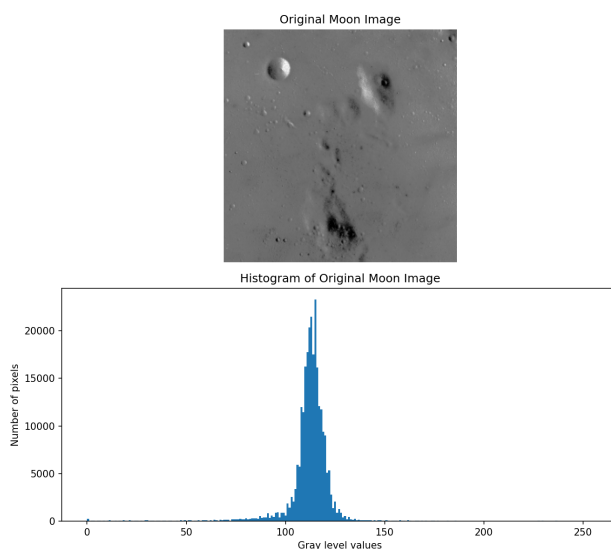


Figure 2. Original moon image and histogram.

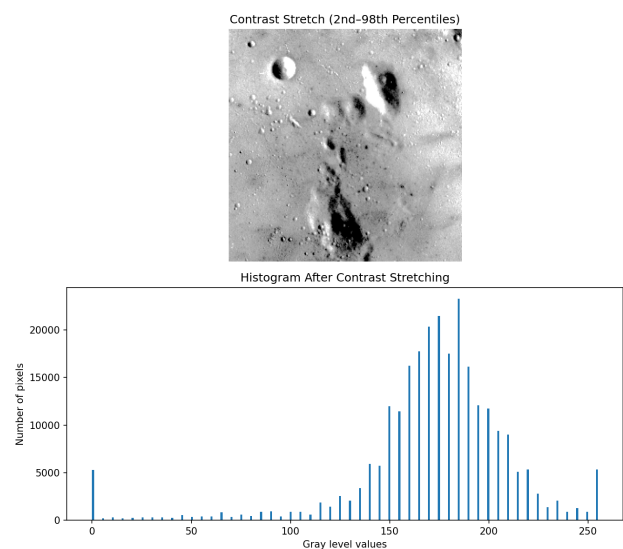


Figure 3. Contrast-stretched moon image and histogram (2nd-98th percentile range).

3. Assessment Task 1 - Contrast Stretching (3rd-80th Percentiles)

The moon image was used again. The 3rd percentile was **87** and the 80th percentile was **118**. The intensities within this interval were rescaled across the available output range. Intensities below the lower percentile and above the upper percentile are clipped by the rescaling operation.

Moon - 3rd to 80th Percentile Stretch



Histogram - Task 1

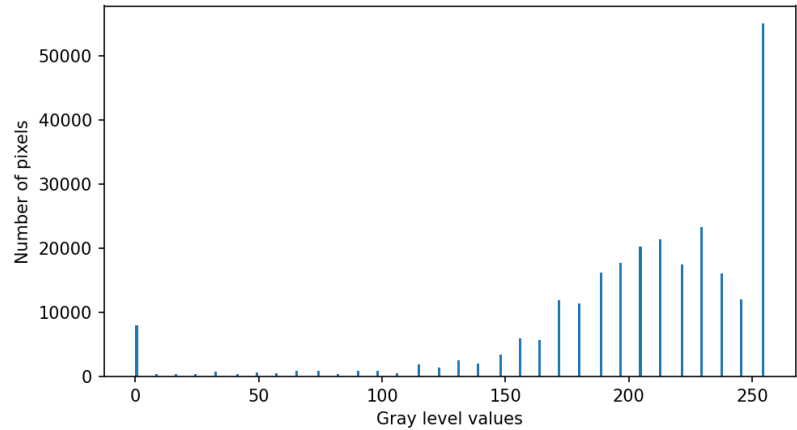


Figure 4. Task 1 result and histogram.

Observation: The selected percentile interval is expanded to improve contrast within the main intensity region of the image. Clipping outside the selected interval produces brighter and darker extremes in the output.

4. Assessment Task 2 - Histogram Equalization

The same moon image was processed using `exposure.equalize_hist`. The function returns a floating-point image with normalized intensity values from approximately 0 to 1. The equalization redistributes intensity values according to the cumulative distribution of the image histogram.

Moon - Histogram Equalization



Histogram - Task 2

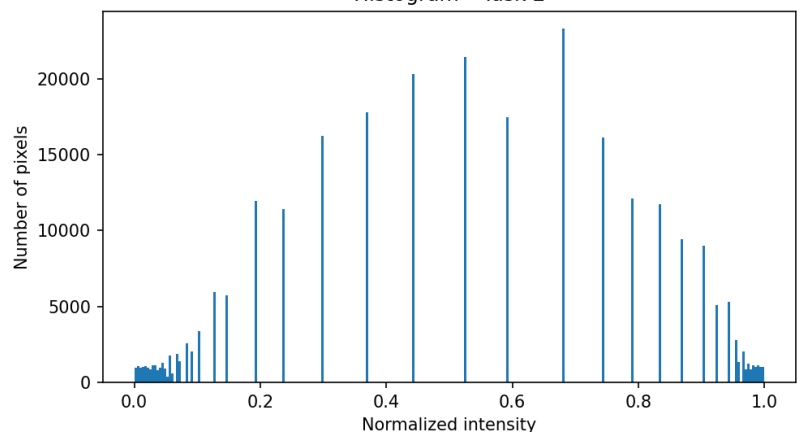


Figure 5. Equalized moon image and histogram.

Observation: Histogram equalization spreads commonly occurring intensity levels over a wider normalized range. This can reveal details in regions that have limited contrast in the original image.

5. Assessment Task 3 - Histogram Matching

Histogram matching was implemented using the scikit-image `match_histograms` function. The Chelsea image was used as the source and the Rocket image as the reference, with RGB channels matched independently using `channel_axis=-1`.



Figure 6. Source (Chelsea), reference (Rocket), and histogram-matched result.

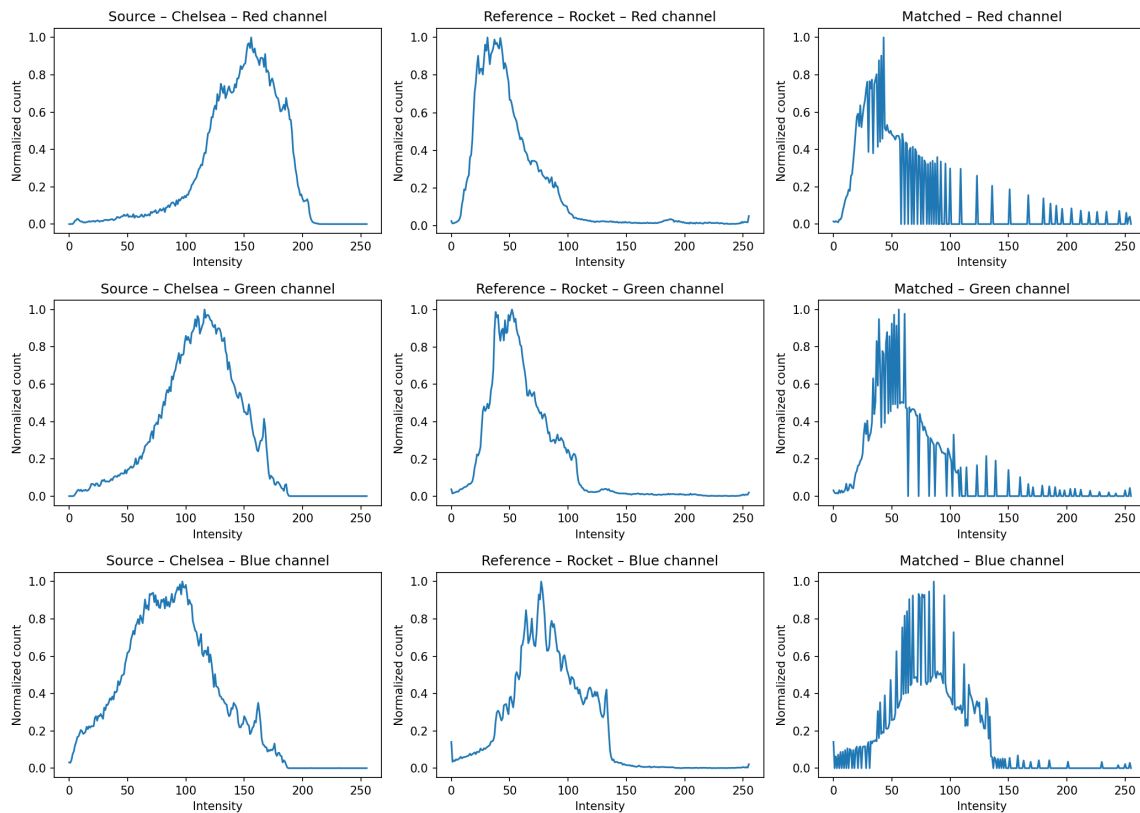


Figure 7. RGB histograms for the source, reference, and matched images.

Observation: After matching, the intensity distribution of Chelsea is adjusted toward the distribution of the Rocket reference image for each RGB channel. The output therefore exhibits intensity and color characteristics closer to the reference distribution.

6. Final Results Summary

All three assessed tasks were completed and the generated results were saved as image files inside the **Lab4_Images** folder used by the notebook.

Task	Operation	Input / Configuration	Result
1	Contrast stretching	moon; 3rd and 80th percentiles	Stretched image + histogram
2	Histogram equalization	moon; exposure.equalize_hist	Equalized image + histogram
3	Histogram matching	Source: Chelsea; Reference: Rocket	Matched image + RGB histograms

7. Resources

1. Student Procedural Manual - ARTI 404, Lab 4.
2. scikit-image - Histogram Equalization:
https://scikit-image.org/docs/stable/auto_examples/color_exposure/plot_equalize.html
3. scikit-image - Histogram Matching:
https://scikit-image.org/docs/0.24.x/auto_examples/color_exposure/plot_histogram_matching.html

8. Submission Contents

The lab submission package contains the completed Jupyter notebook and the generated/used images required for the Lab 4 folder.

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